RESQCORE TECHNICAL SPECIFICATION · V3.2

Database Architecture & Real-Time Platform Synchronization

Complete engineering design specification for end-to-end integration across Citizen Mobile Apps, Tactical Rescue Team Devices, and Central Command Portals.

This specification covers data models, PostGIS relational schemas, event-driven WebSockets/SSE pipelines, store-and-forward offline resilience, and deployment configurations for mission-critical disaster management operations.

PLATFORM ECOSYSTEM
ResQCore Unified Disaster Platform

PRIMARY DATABASE
PostgreSQL 16 + PostGIS Spatial Engine

REAL-TIME BACKBONE
Redis Pub/Sub · WebSocket Streams · SSE

CLIENT IMPLEMENTATIONS
Flutter Mobile (iOS/Android) & React/Vite Web

OPERATIONAL SCENARIOS
Uttarakhand & Kerala Flash Floods, SAR Operations

DOCUMENT STATUS
Production Blueprint & Engineering Guide

CONFIDENTIAL & PROPRIETARY · RESQCORE SYSTEMS

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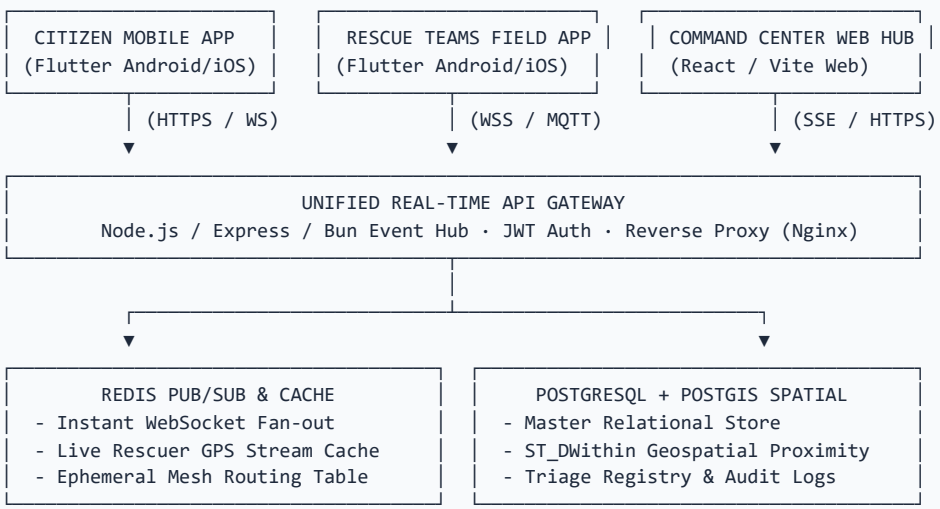
1. Executive Summary & Architectural Overview

In high-stakes disaster scenarios such as flash floods, landslides, and collapsed structures, seconds determine survival. The **ResQCore Disaster Management Platform** bridges the critical communication divide between trapped civilians, rapid-response rescue teams, and centralized tactical dispatch commanders.

The ecosystem consists of three unified clients synchronized over a resilient, event-driven backend:

- **Citizen Mobile App (Flutter)**: A lightweight, intuitive emergency interface allowing victims to trigger 1-Touch SOS alerts, request food/medical supplies, view real-time flood warnings, and track approaching rescue units.
- **Rescue Teams Mobile App (Flutter)**: A ruggedized tactical field application providing incident dispatch alerts, offline topographic maps, GPS member telemetry, digital casualty triage, and satellite mesh sync.
- **Central Command Center (React/Vite Web)**: A high-density tactical web console providing incident triage queues, drone/boat live-tracking radar, relief camp capacity monitors, and emergency broadcast dispatchers.

Platform Topology Diagram



> **Key Architectural Tenet: The 300ms Rule**
>
> Every SOS trigger emitted by a citizen must reflect on the Command Center radar and alert the nearest rescue boat within 300 milliseconds over cellular networks, and degrade gracefully into peer-to-peer radio/mesh mode during cellular blackout.

2. Database Architecture & PostGIS Spatial Design

A conventional relational database is inadequate for disaster operations because incident dispatch requires continuous **spatial computation** (calculating distances on spherical Earth geometry, testing polygon containment inside flood hazard zones, and radius-based bounding box queries).

ResQCore adopts **PostgreSQL 16** equipped with the **PostGIS** extension using standard `WGS 84 (SRID 4326)` GPS coordinates.

2.1 Complete Relational DDL Schema

```
-- Enable UUID Generation and PostGIS Spatial Engine
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "postgis";

-- 1. USERS & ROLES
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    role VARCHAR(20) NOT NULL CHECK (role IN ('citizen', 'rescuer', 'commander', 'dispatcher')),
    full_name VARCHAR(120) NOT NULL,
    phone_number VARCHAR(20) UNIQUE NOT NULL,
    email VARCHAR(120) UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    blood_group VARCHAR(5),
    emergency_contact_name VARCHAR(120),
    emergency_contact_phone VARCHAR(20),
    home_address TEXT,
    sector VARCHAR(100),
    medical_notes TEXT,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- 2. EMERGENCY INCIDENTS & CITIZEN SOS
CREATE TABLE incidents (
    id VARCHAR(30) PRIMARY KEY, -- e.g. 'RQ-261013-4821'
    citizen_id UUID REFERENCES users(id) ON DELETE SET NULL,
    citizen_name VARCHAR(120) NOT NULL,
    contact_phone VARCHAR(20) NOT NULL,
    title VARCHAR(150) NOT NULL,
    type VARCHAR(50) NOT NULL CHECK (type IN ('Flood', 'Landslide', 'Medical', 'Structural', 'Fire', 'General')),
    severity VARCHAR(20) NOT NULL CHECK (severity IN ('Critical', 'High', 'Medium', 'Low')),
    status VARCHAR(30) NOT NULL DEFAULT 'pending'
        CHECK (status IN ('pending', 'triage', 'assigned', 'en_route', 'on_site', 'resolved', 'cancelled')),
    description TEXT,
    location_label VARCHAR(255) NOT NULL,
    -- PostGIS Point Geometry: (Longitude, Latitude) in SRID 4326
    location_geom GEOMETRY(Point, 4326) NOT NULL,
    assigned_team_id UUID,
    people_affected INT DEFAULT 1,
    medical_urgency BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- Fast Spatial & Status Indexes
CREATE INDEX idx_incidents_spatial ON incidents USING GIST (location_geom);
CREATE INDEX idx_incidents_status ON incidents (status);
CREATE INDEX idx_incidents_severity ON incidents (severity);

-- 3. RESCUE TEAMS & TACTICAL UNITS
CREATE TABLE rescue_teams (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    callsign VARCHAR(80) NOT NULL UNIQUE, -- e.g. 'Garhwal SAR Alpha-7'
    unit_type VARCHAR(50) NOT NULL CHECK (unit_type IN ('Swiftwater Boat', 'Alpine Helicopter', 'Canine Search',
    sector VARCHAR(100) NOT NULL,
    status VARCHAR(30) DEFAULT 'ready' CHECK (status IN ('ready', 'dispatched', 'on_mission', 'rest', 'offline')
    commander_id UUID REFERENCES users(id),
    radio_frequency VARCHAR(40),
    active_incident_id VARCHAR(30) REFERENCES incidents(id),
```

```
    current_location GEOMETRY(Point, 4326),
    battery_level INT DEFAULT 100,
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_rescue_teams_spatial ON rescue_teams USING GIST (current_location);

-- 4. HIGH-FREQUENCY RESCUE TELEMETRY
CREATE TABLE team_telemetry (
    id BIGSERIAL PRIMARY KEY,
    team_id UUID NOT NULL REFERENCES rescue_teams(id) ON DELETE CASCADE,
    rescuer_id UUID REFERENCES users(id),
    location GEOMETRY(Point, 4326) NOT NULL,
    altitude_meters FLOAT,
    speed_kmh FLOAT,
    heading_degrees FLOAT,
    battery_percentage INT,
    recorded_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_telemetry_team_time ON team_telemetry (team_id, recorded_at DESC);

-- 5. RELIEF SHELTERS & CASUALTY REGISTRY
CREATE TABLE shelters (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(150) NOT NULL,
    sector VARCHAR(100) NOT NULL,
    capacity INT NOT NULL,
    current_occupancy INT DEFAULT 0,
    food_status VARCHAR(20) DEFAULT 'Adequate',
    medical_supplies_status VARCHAR(20) DEFAULT 'Adequate',
    location GEOMETRY(Point, 4326) NOT NULL
);

CREATE TABLE casualty_registry (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    full_name VARCHAR(120) NOT NULL,
    age INT,
    gender VARCHAR(20),
    blood_group VARCHAR(5),
    status VARCHAR(30) NOT NULL CHECK (status IN ('missing', 'rescued', 'admitted', 'safe', 'deceased')),
    shelter_id UUID REFERENCES shelters(id),
    last_known_location TEXT,
    medical_notes TEXT,
    triage_tag VARCHAR(20) CHECK (triage_tag IN ('Red_Immediate', 'Yellow_Delayed', 'Green_Minor', 'Black_Deceas
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- 6. DYNAMIC HAZARD ZONES (Flood Inundation & Landslide Polygons)
CREATE TABLE hazard_zones (
    id VARCHAR(50) PRIMARY KEY, -- e.g. 'ZONE-CHAMOLI-FLOOD-01'
    zone_name VARCHAR(120) NOT NULL,
    hazard_type VARCHAR(50) NOT NULL, -- 'Flood', 'Landslide', 'Bridge Collapse'
    risk_level VARCHAR(20) NOT NULL CHECK (risk_level IN ('Extreme', 'High', 'Moderate', 'Watch')),
    boundary_geom GEOMETRY(Polygon, 4326) NOT NULL,
    evacuation_ordered BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_hazard_boundary ON hazard_zones USING GIST (boundary_geom);
```

3. Geospatial Querying & Tactical Dispatch Engine

Using PostGIS spatial indexing (R-Tree based `GIST` index), the platform performs spatial operations in under 2 milliseconds across tens of thousands of coordinates.

3.1 Nearest Rescue Team Dispatch Algorithm

When a citizen files an SOS, the system automatically finds the 3 nearest ready rescue units, factoring in Euclidean geographic distance:

```
-- Find the 3 closest available rescue units to an SOS location (Lng: 78.0322, Lat: 30.3165)
SELECT
    t.id AS team_id,
    t.callsign,
    t.unit_type,
    t.sector,
    ROUND(ST_Distance(
        t.current_location::geography,
        ST_SetSRID(ST_Point(78.0322, 30.3165), 4326)::geography
    )::numeric, 0) AS distance_meters
FROM rescue_teams t
WHERE t.status = 'ready'
ORDER BY t.current_location <-> ST_SetSRID(ST_Point(78.0322, 30.3165), 4326)
LIMIT 3;
```

3.2 Polygon Geofencing: Immediate Evacuation Alerts

When a river breaks its banks, commanders draw or upload a flood hazard polygon. The database immediately retrieves all registered citizens residing inside that polygon to trigger mass evacuation SMS and in-app alerts:

```
-- Identify all citizens residing inside a dangerous inundation polygon
SELECT
    u.id,
    u.full_name,
    u.phone_number,
    u.home_address
FROM users u
JOIN hazard_zones hz ON hz.id = 'ZONE-CHAMOLI-FLOOD-01'
WHERE ST_Contains(
    hz.boundary_geom,
    u.home_geom
)
AND u.role = 'citizen';
```

4. Real-Time Synchronization Engine

To maintain real-time responsiveness without polling lag, ResQCore deploys a **Multi-Protocol Event Engine** combining Server-Sent Events (SSE) for web dashboards, WebSockets for high-frequency mobile streams, and Redis Pub/Sub for distributed scaling.

4.1 End-to-End Real-Time Event Matrix

Event Name	Publisher	Protocol	Subscribers	Payload Summary
`new_citizen_sos`	Citizen App	REST / WSS	Command Center, Assigned Teams	Incident ID, GPS coords, severity, medical notes
`incident_assigned`	Command Center	REST -> Redis	Citizen App, Rescue Team App	Assigned team callsign, ETA, frequency
`rescuer_telemetry_ping`	Rescue App	WSS (every 3s)	Command Center Live Radar, Citizen	Team ID, Lat/Lng, Heading, Altitude, Battery
`hazard_zone_alert`	Command Center	SSE / FCM	All Citizens in Geofence	Zone polygon, evacuation shelter directions
`casualty_status_update`	Rescue Triage	REST / WSS	Command Center, Shelter Staff	Victim ID, triage tag (Red/Yellow/Green), destination

4.2 Node.js Central Real-Time Hub Implementation

Below is the production implementation of the real-time event pipeline linking the REST API, Redis Pub/Sub, and WebSocket connections:

```
// server/realtimeHub.js
import express from 'express';
import { createServer } from 'http';
import { WebSocketServer } from 'ws';
import Redis from 'ioredis';
import pg from 'pg';

const app = express();
const server = createServer(app);
const wss = new WebSocketServer({ server });
const redisPub = new Redis(process.env.REDIS_URL);
const redisSub = new Redis(process.env.REDIS_URL);
const pool = new pg.Pool({ connectionString: process.env.DATABASE_URL });

app.use(express.json());

// 1. Citizen Emergency SOS Ingestion Endpoint
app.post('/api/citizen/sos', async (req, res) => {
  const { citizenId, citizenName, phone, title, type, severity, lat, lng, description } = req.body;
  const incidentId = `RQ-${Date.now().toString().slice(-6)}`;

  // Spatial Insert with PostGIS
  const query = `
    INSERT INTO incidents (id, citizen_id, citizen_name, contact_phone, title, type, severity, location_label, lo
    VALUES ($1, $2, $3, $4, $5, $6, $7, $8, ST_SetSRID(ST_Point($9, $10), 4326), $11)
    RETURNING *;
  `;
  const result = await pool.query(query, [incidentId, citizenId, citizenName, phone, title, type, severity, 'Live
  const newIncident = result.rows[0];

  // Publish to Redis Channel
  await redisPub.publish('disaster:events', JSON.stringify({
    event: 'new_citizen_sos',
    data: newIncident
  }));

  res.status(201).json({ success: true, incident: newIncident });
});

// 2. Redis Subscription -> WebSocket Broadcast to Web & Field Apps
redisSub.subscribe('disaster:events', 'disaster:telemetry');
redisSub.on('message', (channel, message) => {
  const payload = JSON.parse(message);
  wss.clients.forEach(client => {
    if (client.readyState === 1) { // OPEN
      client.send(JSON.stringify(payload));
    }
  });
});

// 3. High-Frequency Rescuer GPS Stream
wss.on('connection', (ws) => {
  ws.on('message', async (raw) => {
    const msg = JSON.parse(raw);
    if (msg.type === 'telemetry') {
      // Store in Redis ephemeral cache (expires in 60s)
      await redisPub.hset(`telemetry:team:${msg.teamId}`, {
        lat: msg.lat,
        lng: msg.lng,
        battery: msg.battery,
        updatedAt: Date.now()
      });
      // Broadcast live movement pin to Command Center
```

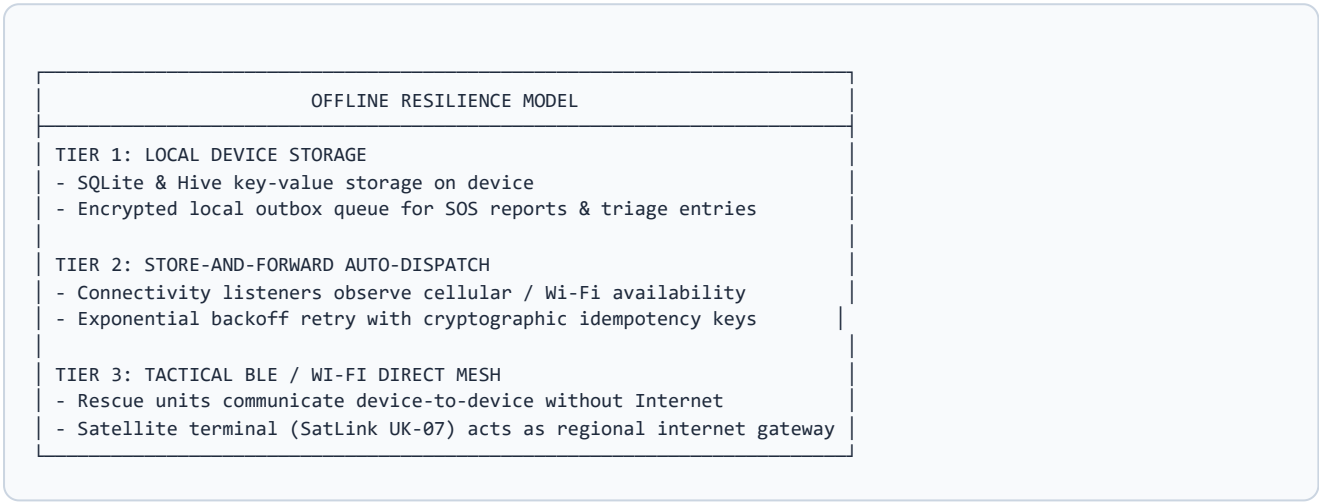
```
      redisPub.publish('disaster:telemetry', JSON.stringify(msg));
    }
  });
});

server.listen(5000, () => console.log('ResQCore Real-Time Hub running on port 5000'));
```

5. Disaster Zone Offline-First Resilience

When disaster strikes, telecommunications infrastructure is frequently the first casualty: fiber lines snap, cellular towers lose power, and networks saturate. ResQCore employs an **Offline-First Resilience Architecture**:

5.1 The 3-Tier Sync Strategy



5.2 Flutter Local Storage & Outbox Implementation

Both the Citizen App and Rescue Teams App maintain an offline SQLite database. If a citizen taps "Instant SOS" during an internet outage:

1. The request is given a local UUID and saved to `offline_sos_queue` with timestamp and GPS coords.
2. The app UI shows "*Queued Offline · Retrying automatically*" so the user remains calm.
3. A background work manager detects cellular signal restoration and POSTs the queue to the backend.
4. The server acknowledges and the local outbox is cleared.

6. Security, Authentication & Role-Based Access Control

User Role	App Interface	Auth Mechanism	Permissions & Access Control
Citizen	Citizen Mobile App	Phone/Email + Password (BCrypt) + JWT	Submit personal SOS; view assigned rescue status; view regional public alerts and relief camps.
Rescue Operator	Rescue Teams App	Operator ID + Mission Key + Biometric	Stream GPS telemetry; update casualty triage status; report field obstacles; join tactical voice/radio.
Incident Commander	Command Center Web	MFA + Hardware Key / RSA Token	Dispatch rescue teams; declare evacuation zones; issue district alerts; override triage classifications.

Security & Encryption Standards

- **Passwords**: Hashed using BCrypt with minimum salt factor 12.
- **Transit Security**: All WebSockets and HTTP calls enforce TLS 1.3 / WSS.
- **Telemetry Privacy**: Civilian health and blood group data are encrypted at rest and disclosed only to verified first responders actively assigned to that civilian's incident.

7. Complete Deployment & Production Setup Blueprint

7.1 Production Environment Variables (.env)

```
# Database Configuration (PostgreSQL + PostGIS)
DATABASE_URL=postgresql://resq_admin:StrongPass2026!@postgres.internal:5432/resqcore_db?sslmode=require

# In-Memory Cache & Pub/Sub
REDIS_URL=redis://:RedisSecureKey2026@redis.internal:6379

# Server Ports & CORS
PORT=5000
CORS_ORIGIN=https://command.resqcore.org,http://localhost:5173

# Security & Tokens
JWT_SECRET=c8f8b89e273bc92841029487cba10293847561029384756
ENCRYPTION_KEY=32_byte_secure_hex_encryption_key_here

# Telephony Gateway (When Activated)
TWILIO_ACCOUNT_SID=AC_dummy_or_active_key
TWILIO_AUTH_TOKEN=auth_token_here
TWILIO_PHONE_NUMBER=+18005550199
```

7.2 Docker Compose Production Deployment

```
version: '3.8'

services:
  postgres:
    image: postgis/postgis:16-3.4
    container_name: resqcore_postgres
    restart: always
    environment:
      POSTGRES_USER: resq_admin
      POSTGRES_PASSWORD: StrongPass2026!
      POSTGRES_DB: resqcore_db
    ports:
      - "5432:5432"
    volumes:
      - pgdata:/var/lib/postgresql/data

  redis:
    image: redis:7-alpine
    container_name: resqcore_redis
    restart: always
    command: redis-server --requirepass RedisSecureKey2026
    ports:
      - "6379:6379"

  api_gateway:
    build: ./resqcore
    container_name: resqcore_api
    restart: always
    ports:
      - "5000:5000"
    environment:
      - DATABASE_URL=postgresql://resq_admin:StrongPass2026!@postgres:5432/resqcore_db
      - REDIS_URL=redis://:RedisSecureKey2026@redis:6379
    depends_on:
      - postgres
      - redis

volumes:
  pgdata:
```